

30 Practical Computer Assignments (Hands-On Only)

1. Basic Computer Use

- 1  Computer ON / OFF सही तरीके से करो
- 2  Mouse से Right-click, Left-click, Scroll practice करो
- 3  Keyboard से 10 lines typing करो (Notepad में)
- 4  Date & Time check और change करो
- 5  Screen resolution change करके वापस normal करो

File & Folder Practice

- 6  Desktop पर 5 folders बनाओ
- 7  Folder rename करो
- 8  एक folder delete करो
- 9  Recycle Bin से file restore करो
- 10  Notepad में file बनाकर save करो

System Information

11. System Properties खोलो (RAM, OS देखो)

- 12  Task Manager खोलकर running apps देखो
- 13  Disk space check करो (C, D drive)
- 14  Volume, Sound settings change करो
- 15  Sleep, Restart options use करो

Internet Practice

- 16  Chrome / Edge खोलो
- 17  Google पर कुछ search करो
- 18  Website bookmark करो
- 19  Image download करके folder में save करो
- 20  Image upload करके email में attach करो

Email & Cloud

- 21  Gmail login करो
- 22  New email compose करके भेजो
- 23  File attach करके email भेजो
- 24  Google Drive में file upload करो
- 25  Drive से share link बनाओ

2. System & OS Advanced

- 1  System Configuration (msconfig) खोलकर startup apps check करो
- 2  Task Manager → Startup programs enable/disable करो
- 3  New User Account बनाओ (Admin vs Standard)
- 4  Windows password change + lock screen customize करो

5 System Restore point create करो

♦ File, Disk & Storage

6 Hidden files enable/disable करो

7 Disk Cleanup + Disk Optimization (Defragment) चलाओ

8 Disk Management खोलकर partitions देखो

9 ZIP / UNZIP files बनाओ

10 Advanced file search (date, size, type)

♦ Software & Installation

11 Software safely download & install करो

12 Unwanted software completely uninstall करो

13 Software update manually check करो

14 Default apps change करो

15 Portable software use करो

♦ Networking & Internet (Advanced Use)

16 IP address check करो (ipconfig)

17 Wi-Fi network details देखो

18 Router login (demo) समझो

19 Firewall ON/OFF check करो

20 Upload vs Download speed test करो

♦ Security & Safety

21 Antivirus scan run करो

22 Folder password / encryption करो

23 Phishing email identify करो

24 Strong password manager use करो

25 Windows Defender settings explore करो

3. EXPERT / PROFESSIONAL PRACTICAL TASKS

 1. Operating System (Pro Level)

1 Event Viewer खोलकर system errors check करो

2 Windows Services में services start/stop करो

3 Group Policy Editor explore करो (gpedit.msc)

4 Environment Variables देखो और edit समझो

5 Safe Mode में system boot करना सीखो

 2. Disk, Backup & Recovery

6 Full system backup create करो

7 Recovery Drive / Bootable USB बनाओ

- 8] File History enable करो
- 9] Disk error check (CHKDSK) run करो
- 10] Restore backup करके test करो

3. Networking (Hands-on)

- 11] Static IP set करना सीखो
- 12] Ping, Tracert, Netstat commands run करो
- 13] Network sharing enable/disable करो
- 14] Proxy settings समझो
- 15] Remote Desktop connection use करो

4. Security (Professional Level)

- 16] BitLocker encryption enable/disable करो
- 17] Windows Defender advanced scan schedule करो
- 18] User permissions (Read / Write / Full control) set करो
- 19] Fake website पहचानने की practice करो
- 20] Audit logs check करो

5. Command Line & Power Tools

- 21] Command Prompt में 10 commands practice करो (dir, cd, copy, move, del, ipconfig, ping, net user...)
- 22] Batch file (.bat) बनाकर task automate करो
- 23] PowerShell open करके basic commands run करो
- 24] System info export करना सीखो
- 25] Shutdown / restart schedule करो

6. Cloud & Virtualization

- 26] Google Drive / OneDrive sync setup करो
- 27] Virtual Machine (VirtualBox) install करो
- 28] Virtual OS (Windows/Linux) run करो
- 29] Snapshot create & restore करो
- 30] Cloud + VM comparison report बनाओ

4 . ULTRA ADVANCED PRACTICAL TASKS

1. Operating System – ADMIN LEVEL

- 1] Local Security Policy खोलकर password rules set करो
- 2] Multiple user login test करो (permission difference)
- 3] Windows Registry Editor खोलकर read-only keys समझो
- 4] Startup delay issue identify & fix करो
- 5] System logs से error code निकालकर Google-fix करो

2. Disk, File System & Permissions

- 6 NTFS permissions (Read/Write/Modify) apply करो
- 7 Shared folder बनाकर network access control करो
- 8 Disk quota set करो (user storage limit)
- 9 Corrupt file recovery try करो
- 10 Shadow Copy (Previous Versions) restore करो

3. Networking – REAL COMMAND PRACTICE

- 11 Local network map बनाओ (IP + device list)
- 12 DNS flush + renew IP करो
- 13 Port number identify करो (80, 443, 21, 22)
- 14 Network issue simulate करो (disable adapter → fix)
- 15 Remote PC troubleshoot using Remote Desktop

4. Cyber & System Security (PRACTICAL)

- 16 Fake email / phishing site manually identify करो
- 17 File hash check करो (certutil)
- 18 Secure folder with encryption + permission
- 19 USB security policy test करो
- 20 Firewall inbound/outbound rule create करो

5. Command Line + Automation

- 21 Advanced CMD commands (tasklist, taskkill, netstat)
- 22 Batch script:
 - ➔ auto-create folders + files
- 23 PowerShell script run करके system info export करो
- 24 Auto-shutdown script test करो
- 25 Log file create करके daily activity save करो

6. Virtualization + Cloud (PRO LEVEL)

- 26 VM में networking mode change (NAT / Bridged)
- 27 VM snapshot failure → restore scenario
- 28 Linux VM basic commands run करो
- 29 Cloud folder sync conflict resolve करो
- 30 Cloud backup + local restore test करो

5. EXTREME ADVANCED PRACTICAL LABS

1. OS INTERNALS (ENTERPRISE LEVEL)

- 1 Windows Boot Process trace करो (Boot log enable)
- 2 BSOD error code identify + root cause analysis करो
- 3 Event Viewer से critical error → solution report बनाओ
- 4 User Profile corruption simulate + fix करो

5 Windows performance bottleneck identify (CPU/RAM/Disk)

2. Storage, File System & Recovery (REAL SCENARIOS)

- 6 Disk failure scenario simulate (read-only / offline disk)
- 7 NTFS inheritance break करके custom permissions लगाओ
- 8 Deleted file recovery using Windows tools
- 9 VSS (Volume Shadow Copy) manually trigger करो
- 10 Storage optimization report बनाओ (before/after)

3. ADVANCED NETWORK OPERATIONS

- 11 Complete network troubleshooting flow execute करो
- 12 ARP table check + flush करो
- 13 Port conflict identify करो (netstat + taskkill)
- 14 DNS issue simulate → resolve
- 15 Network latency diagnose (ping + tracert)

4. SECURITY & INCIDENT RESPONSE

- 16 Malware behavior identify (safe demo file)
- 17 Windows Defender logs analyze करो
- 18 Privilege escalation risk identify करो
- 19 USB attack prevention policy implement करो
- 20 Incident response report तैयार करो

5. AUTOMATION & ADMIN TOOLS

- 21 Advanced PowerShell script (user + folder automation)
- 22 Scheduled task create (daily backup script)
- 23 System audit script generate करो
- 24 Log rotation simulate करो
- 25 Admin tool documentation बनाओ

6. VIRTUALIZATION + HYBRID CLOUD

- 26 Multi-VM network lab create करो
- 27 VM resource starvation → fix
- 28 Linux VM + Windows VM communication test
- 29 Cloud sync failure scenario resolve
- 30 Hybrid backup strategy implement करो

6. ULTIMATE ADVANCED PRACTICAL LABS

1. Identity & Access (Enterprise Core)

- 1 Local users vs Groups policy design करो
- 2 Password policy + account lockout test करो

- 3 File server permissions (Role-based access) लागू करो
- 4 Login audit (who logged in/failed) निकालो
- 5 Least-Privilege model implement करो

2. Active Directory (Simulation)

- 6 AD DS lab (VM में) setup करो
- 7 OU structure बनाओ (HR, IT, Admin)
- 8 User lifecycle: Create → Disable → Delete
- 9 Group Policy (GPO) से USB block करो
- 10 AD backup + restore test करो

3. Linux Fundamentals (Admin-Use)

- 11 Linux install (VM) + user create
- 12 Permissions (chmod, chown) practice
- 13 Services start/stop (systemctl)
- 14 Logs analyze (/var/log)
- 15 Bash script: backup + cleanup

4. Networking (Ops Level)

- 16 Subnetting practice (real IP plan)
- 17 VLAN concept lab (simulated)
- 18 NAT vs Bridged vs Host-only (VM)
- 19 DNS records test (A, CNAME)
- 20 Packet loss & latency RCA report

5. Security Operations (SOC-Ready)

- 21 Baseline hardening checklist लागू करो
- 22 Vulnerability scan (safe tool) + report
- 23 Log correlation (Security + System)
- 24 Incident triage: detect → contain → report
- 25 Backup integrity + ransomware drill (mock)

6. Automation & Ops

- 26 PowerShell: bulk user + folder creation
- 27 Scheduled tasks: daily backup + log
- 28 Health-check script (CPU/RAM/Disk)
- 29 Email alert on threshold breach
- 30 Ops runbook documentation बनाओ

7. Cloud & Hybrid Ops

- 31 Cloud storage lifecycle policy
- 32 IAM roles & permissions test

- 33 On-prem ↔ Cloud sync conflict resolve
- 34 Cost-optimization checklist
- 35 DR drill: local fail → cloud restore

7. 🦋 ABSOLUTE ADVANCED – REAL PRODUCTION LABS

🧠 1. SYSTEM FAILURE & ROOT CAUSE (RCA)

- 1 System crash simulate करो (safe way)
- 2 Boot failure log analyze करो
- 3 CPU 100% issue → root cause निकालो
- 4 Memory leak process identify करो
- 5 RCA document बनाओ (Problem → Cause → Fix)

💾 2. ENTERPRISE STORAGE & DATA PROTECTION

- 6 RAID types (0/1/5) simulate (VM)
- 7 Data corruption scenario → recovery attempt
- 8 Backup vs Snapshot difference test
- 9 Immutable backup concept implement
- 10 Data loss post-mortem report बनाओ

🌐 3. NETWORK INCIDENT WAR-ROOM

- 11 Complete network outage scenario simulate
- 12 Switch vs Router issue differentiate
- 13 DNS poisoning concept analyze
- 14 Packet capture (concept + demo)
- 15 Network incident timeline बनाओ

🔒 4. SECURITY BREACH & RESPONSE

- 16 Unauthorized access attempt detect करो
- 17 Privilege misuse case analyze
- 18 Malware entry point identify करो
- 19 Containment + eradication steps लिखो
- 20 Final incident response report बनाओ

⚙️ 5. AUTOMATION = REAL POWER

- 21 Full PC health-check automation
- 22 Auto-backup + auto-cleanup script
- 23 User onboarding automation (account + folder)
- 24 Alerting system (log → action)
- 25 Automation documentation (SOP)

🐧 6. WINDOWS + LINUX CO-EXISTENCE

- 26 Windows ↔ Linux file sharing setup
- 27 Linux server hardening checklist
- 28 Cron jobs vs Task Scheduler comparison
- 29 Cross-platform backup strategy
- 30 Mixed-OS troubleshooting report

7. CLOUD FAILURE & COST CONTROL

- 31 Cloud outage scenario analysis
- 32 Access misconfiguration → fix
- 33 Cost spike root cause analysis
- 34 Backup restore from cloud to local
- 35 Cloud DR (Disaster Recovery) drill

8. GOD-MODE PRACTICAL LABS (NO MERCY)

1. SYSTEM DESIGN & ARCHITECTURE

- 1 Single PC → Office → Enterprise architecture draw करो
- 2 Bottleneck prediction करो (CPU/RAM/Disk/Network)
- 3 SPOF (Single Point of Failure) identify करो
- 4 High-availability concept simulate करो
- 5 Scalability plan बनाओ (users × 10)

2. FAILURE ENGINEERING (CHAOS LAB)

- 6 Random service failure simulate
- 7 Wrong permission disaster → recovery
- 8 Accidental delete → full recovery drill
- 9 Update failure rollback plan
- 10 Downtime impact analysis (₹ loss estimate)

3. NETWORK AT SCALE

- 11 IP exhaustion scenario plan
- 12 Subnet redesign without downtime
- 13 Latency vs Bandwidth optimization
- 14 Zero-trust network concept apply
- 15 Network change management log

4. SECURITY BY DESIGN

- 16 Threat modeling (STRIDE style)
- 17 Attack surface reduction checklist
- 18 Insider threat simulation
- 19 Security vs usability trade-off decision
- 20 Executive security summary report

5. AUTOMATION AS A SYSTEM

-  21 "No-human" daily ops design
-  22 Self-healing concept (detect → fix)
-  23 Alert fatigue reduction plan
-  24 Automation failure fallback
-  25 Automation ROI calculation

6. CROSS-PLATFORM DOMINANCE

-  26 Windows + Linux policy alignment
-  27 Centralized logging strategy
-  28 Config drift detection concept
-  29 Platform migration plan
-  30 Mixed-OS incident handling SOP

7. CLOUD + ON-PREM STRATEGY

-  31 Cloud vs On-prem decision matrix
-  32 Vendor lock-in avoidance plan
-  33 Cost governance framework
-  34 Cloud outage business continuity
-  35 Hybrid future roadmap (3 years)